MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-15B/U

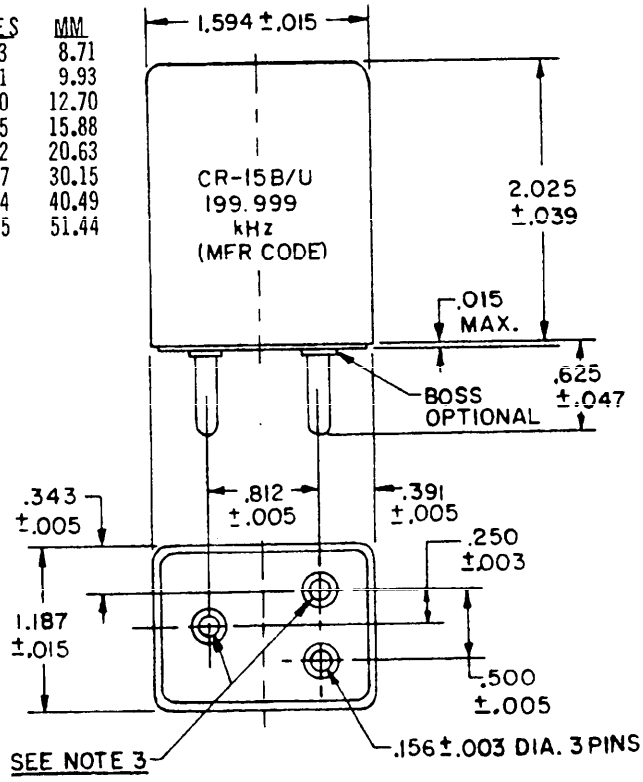
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 80 to 200 kHz; fundamental; noncontrolled; antiresonance.

Ⓓ THIS SPECIFICATION INACTIVE FOR NEW DESIGN

INCHES	MM	INCHES	MM
.003	.08	.343	8.71
.005	.13	.391	9.93
.015	.39	.500	12.70
.039	.99	.625	15.88
.047	1.19	.812	20.63
.156	3.96	1.187	30.15
.250	6.35	1.594	40.49
		2.025	51.44



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The quartz resonator shall be connected between the pins indicated.

FIGURE 1. Dimensions and configuration.

Ⓓ denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
Holder	HC-21/U (MIL-H-10056/10)
Dimensions, marking, and configuration	See figure
Frequency range	80 to 200 kHz, inclusive
(D) Frequency tolerance	±100 PPM
Antiresonance, load capacitance	32.0 ± 0.5 pf
Equivalent resistance:	kHz — Ohms, maximum 80 to 120 — 5,000 120+ to 160 — 5,500 160+ to 200 — 6,000
Mode of oscillation	Fundamental
Operating temperature range	-40° to +70°C, inclusive
Test set calibration values, and rated drive level	See table II
Shock:	
(D) Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
(D) Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
(D) Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength:	kHz — Grams, minimum 80 to 120 — 1,000 120+ to 160 — 800 160+ to 200 — 700

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (0.030 inch total excursion).

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	kHz	Ohms	ma	Volts	mw
TS-710/TSM	80 to 120	3500	---	2.65	2.0 ±0.4
	120+ to 160	3900	---	2.79	
	160+ to 200	4200	---	2.90	

NOTE:

1. This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-1)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/1D CRYSTAL UNIT, QUARTZ, CR-15B/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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C. REASON FOR RECOMMENDED CHANGE(S)

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